

A Pure Analytical Gaussian Expansion Method for Heavy and Mixed-State Quarkonia

Mass spectra and decay widths of $b\bar{b}$, $c\bar{c}$, B_c and D

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1. Motivation & Theoretical Framework

Motivation: Quarkonia as a QCD Laboratory

- **Objective:** Heavy quarkonia — bound states of a heavy quark and its antiquark ($c\bar{c}$, $b\bar{b}$) — are the “hydrogen atom” of the strong interaction: a clean, non-relativistic probe of the **confining, low-energy regime** of QCD where perturbation theory fails.
- **Why non-relativistic:** the heavy quarks are *slow* ($v^2/c^2 \approx 0.3$ for charm, 0.1 for bottom), so a Schrödinger picture with a static potential is well justified.
- **Scope:** a *single* framework spanning **four sectors** — bottomonium $b\bar{b}$, charmonium $c\bar{c}$, and the mixed-flavour $B_c (b\bar{c})$ and $D (c\bar{u})$ — fixed only by the two quark masses and the potential.

The Cornell Potential & Hamiltonian

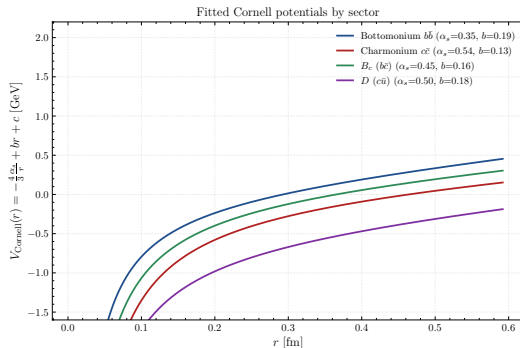
- **Model:** the phenomenological **Cornell potential** — a single-gluon Coulomb attraction plus a *linear* confining term from the QCD string:

$$V(r) = -\frac{4}{3} \frac{\alpha_s}{r} + b r + c$$

- **Full Hamiltonian** adds the spin-dependent Breit–Fermi fine structure:

$$H = \frac{p^2}{2\mu} + V(r) + \underbrace{V_{LS} + V_{SS} + V_T}_{\text{spin terms}}$$

- Short range $\sim$ asymptotic freedom; the linear slope $b \approx 0.18 \text{ GeV}^2$ is the **string tension** — confinement.



The four fitted Cornell potentials.

2. The Gaussian Expansion Method

Methodology: The Gaussian Expansion Method

- **Core technique:** the Rayleigh–Ritz variational method in a basis of **25 Gaussians** $\phi_n(r) = r^l e^{-\nu_n r^2}$, with widths ν_n in *geometric progression* over $r \in [0.05, 15] \text{ GeV}^{-1}$.
- **Generalized eigenproblem:** expanding $\psi = \sum_n c_n \phi_n$ turns $H\psi = E\psi$ into

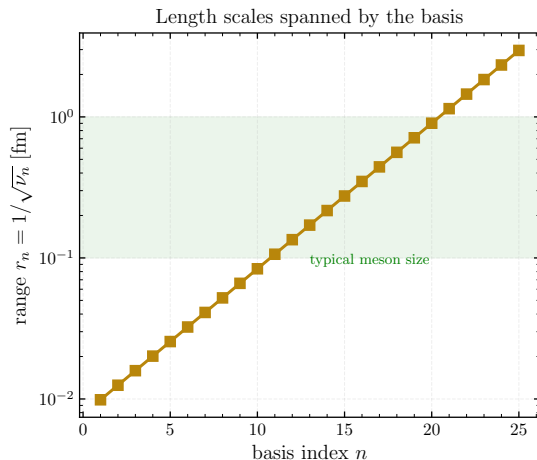
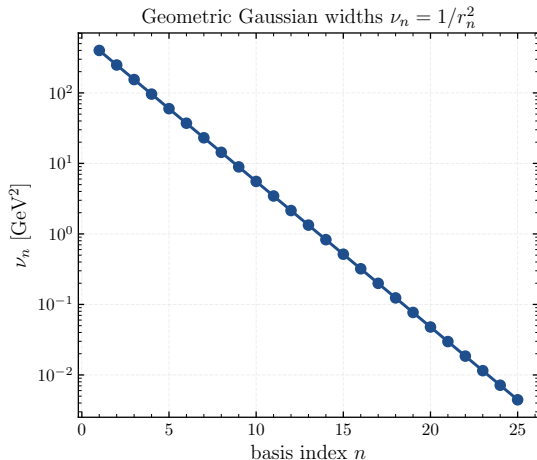
$$H \mathbf{c} = E \mathbf{S} \mathbf{c},$$

(S = overlap), solved with `scipy.linalg.eigh`.

- **Why Gaussians:** one geometric mesh resolves both the short-distance Coulomb cusp and the long confined tail in a single, well-conditioned basis.

The Gaussian Width Mesh

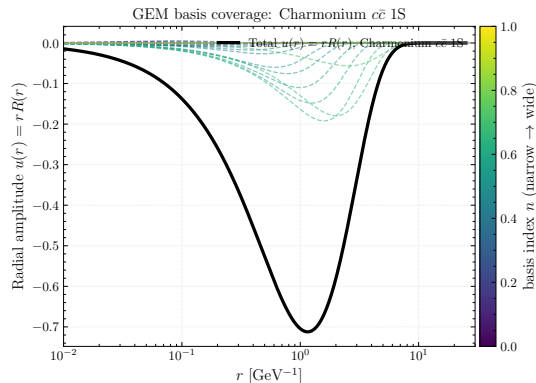
GEM geometric mesh ($n_{\max} = 25$, $r_{\min} = 0.05$, $r_{\max} = 15 \text{ GeV}^{-1}$)



Geometric width mesh $\nu_n = 1/r_n^2$ and the length scales it spans.

The Analytic Engine: Closed-Form Matrix Elements

- **Key trick:** every element of H and S — kinetic, Coulomb, linear, and the Gaussian-smeared hyperfine term — reduces to a closed-form Γ -**function integral**. *No numerical quadrature, no radial grid.*
- **Consequence:** the *only* approximation is the finite basis size. There is no quadrature error to control — the build is fast and **exactly reproducible**.
- This is what “*pure analytical*” means: the GEM here is an algebraic machine, not a numerical integrator.



25 weighted Gaussians (dashed) summing to the charmonium 1S amplitude $u(r) = rR(r)$.

Wavefunctions & Convergence

- **Output:** reconstructed radial amplitudes $u(r)$ show the correct 0/1/2 **radial nodes** for 1S/2S/3S and the tighter binding of the heavier systems.
- **Convergence guarantee (virial theorem):** for a converged basis

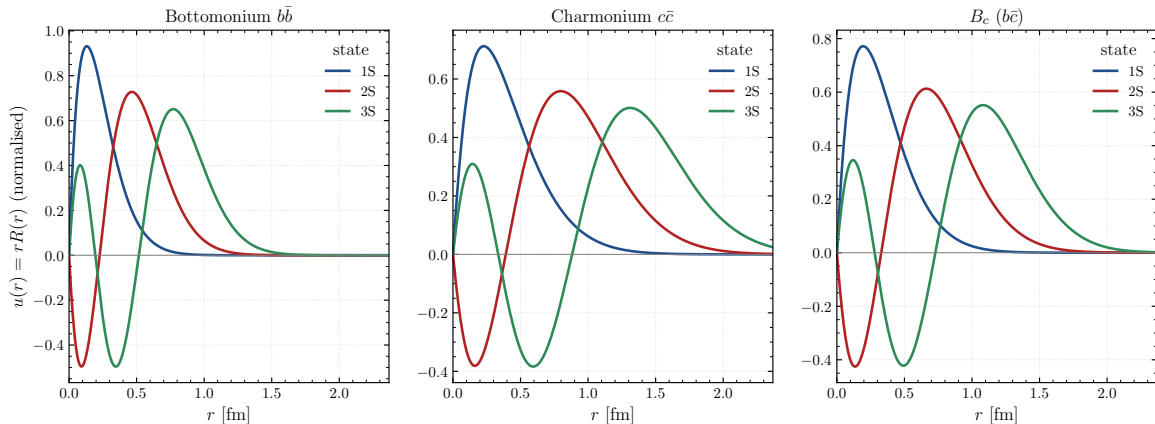
$$\frac{\langle 2T \rangle}{\langle r dV/dr \rangle} = 1.00,$$

including the hyperfine term for S-waves — a *parameter-free* check that the eigenstates are physical.

- If this ratio drifts from 1, the basis is not converged and every downstream number is suspect.

Radial Wavefunctions & Nodes

Reconstructed S-wave radial amplitudes (radial nodes visible)



S-wave amplitudes $u(r) = rR(r)$ for 1S/2S/3S per sector.

3. Spin-Dependent Corrections

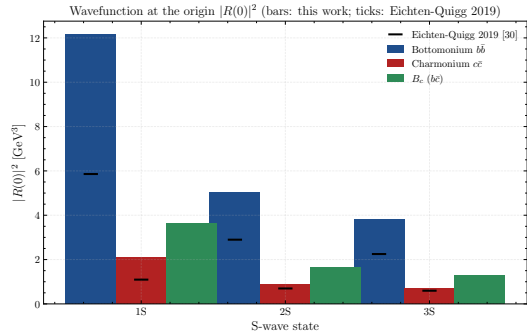
Fine Structure: Breit–Fermi Corrections

Approach: the GEM gives the spin-*averaged* spectrum; the spin splittings are added as **first-order perturbations** — expectation values over the GEM eigenvectors, so they inherit the same analytic matrix elements (no new machinery).

- **Spin–orbit** V_{LS} ($L > 0$): splits the 3P_J and 3D_J multiplets.
- **Spin–spin** V_{SS} (hyperfine): the contact term separating singlets from triplets — e.g. the J/ψ – η_c splitting. It is smeared by a Gaussian of width $\sigma \sim \sqrt{\mu}$ tied to the mass scale.
- **Tensor** V_T : the diagonal tensor splitting *within* the spin-triplet multiplets.

S–D Mixing & the Wavefunction at the Origin

- **Off-diagonal tensor (S–D mixing):** the element $\langle 2^3S_1 | V_T | 1^3D_1 \rangle$ mixes S- and D-waves — indispensable for the $\psi(3770)$, a $2S$ – $1D$ admixture.
- $|R(0)|^2$ **via the hypervirial theorem:** the S-wave wavefunction-at-the-origin — which every annihilation width needs — is obtained as an **expectation value**, never by evaluating ψ at $r = 0$ directly. Far more numerically stable.
- These two ingredients connect the spectrum to the *decay widths* (next sections).



$|R(0)|^2$ for the S-waves — it drives all annihilation widths.

4. Fitting & Uncertainties

Parameter Fitting: A True χ^2 Objective

- **Data:** experimental masses *and* the ground / first-excited leptonic e^+e^- widths, read live from the official **PDG 2026** SQLite tables (no hand-edited inputs).
- **Objective:** a genuinely dimensionless χ^2 — each residual is $(\text{model} - \text{exp})/\sigma_{\text{physical}}$, *not* a hand-tuned weight. Folding the leptonic widths in lets the wavefunction at the origin help pin the potential — masses alone are not enough.
- **Optimizer:** `scipy.optimize.least_squares` with the trust-region-reflective method, which handles the parameter bounds and the under-determined B_c sector that plain Levenberg–Marquardt cannot.

Two Distinct Uncertainties

- **Model uncertainty (the $\pm$ on a value):** finite-difference propagation of the fitted Cornell covariance — “how far does a prediction move when the parameters wiggle within their errors.”
- **Physical error bar (the σ in a χ^2 or pull):**

$$\sigma = \sqrt{\sigma_{\text{exp}}^2 + \sigma_{\text{theory}}^2}, \quad \sigma_{\text{theory}} = 10 \text{ MeV (masses), } 30\% \text{ (widths)}.$$

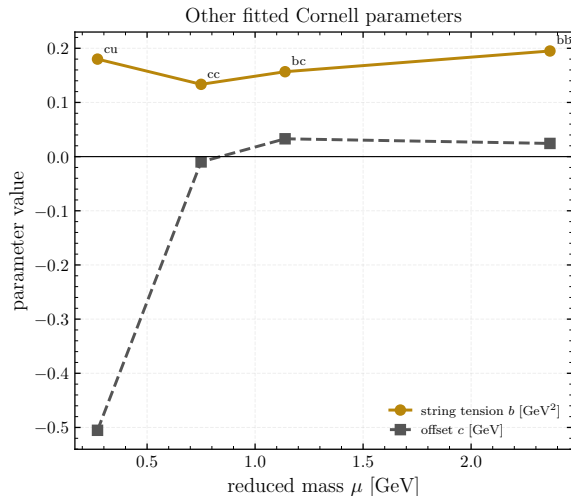
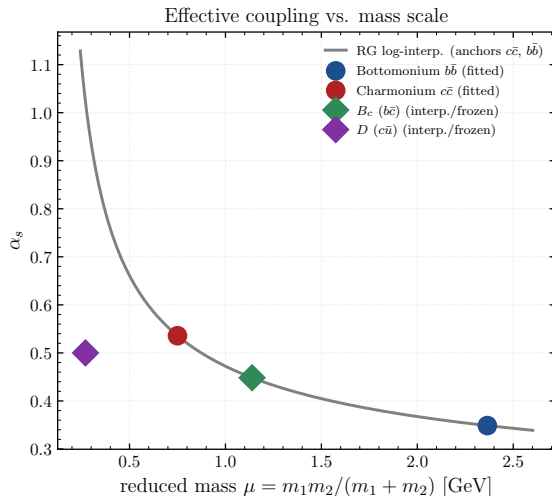
The theory floor represents the physics the NR model omits — relativistic $\mathcal{O}(v^2/c^2)$, coupled channels, quenching.

- **Why it matters:** PDG mass errors are only 0.1–1 MeV; no NR potential model can reach that. Dividing by the honest σ makes $\chi^2/\text{dof} \approx 1$ mean “reproduces the data to within its intrinsic systematic.” The headline figure is the assumption-free **RMS mass deviation**.

Four Sectors & the B_c Interpolation

- $b\bar{b}$ **and** $c\bar{c}$: fully fitted — four free Cornell parameters (α_s, b, c, σ) each.
- B_c **is special**: too few known states to pin α_s and σ . They are **log-interpolated** (running-coupling style) between the $c\bar{c}$ and $b\bar{b}$ fits at the B_c reduced-mass scale, with uncertainties propagated through the interpolation.
- D **meson**: only the offset c is free — a single anchored ground state.

The B_c Coupling by Interpolation



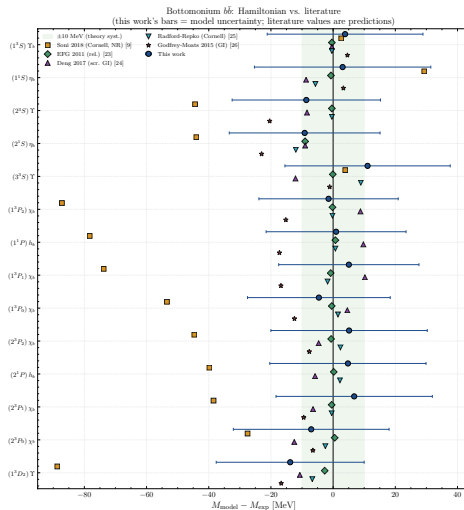
Effective α_s vs reduced-mass scale: $c\bar{c}$ and $b\bar{b}$ anchor the B_c coupling.

5. Results: Mass Spectra

Results: Bottomonium Spectrum

- **RMS deviation 7.0 MeV** ($\chi^2/\text{dof} = 0.67$) — *within* the 10 MeV theory systematic.
- 14 measured levels reproduced; the model then **predicts** the full D-wave tower and the $3S/3P$ levels.
- Largest single tension is the 1^3D_2 at -14 MeV — still sub-percent.

Bottomonium vs. the Literature

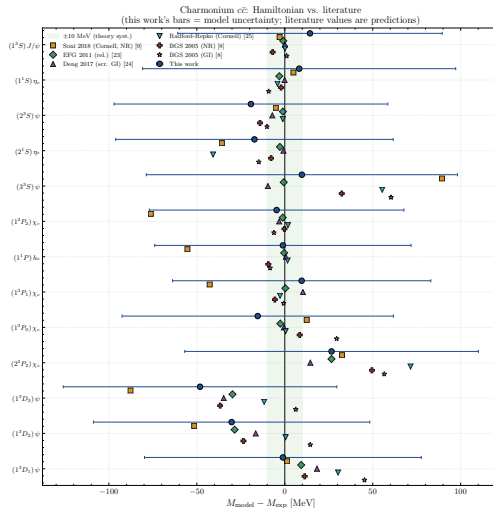


Per-state deviation $M_{\text{model}} - M_{\text{exp}}$: this work vs. five literature models; green band = ± 10 MeV.

Results: Charmonium Spectrum

- **RMS deviation 28.8 MeV** ($\chi^2/\text{dof} = 10.3$) — larger, and *that is physics*.
- Charm is lighter, $v^2/c^2 \approx 0.3$, so the neglected $\mathcal{O}(v^2/c^2)$ corrections exceed the floor — concentrated in the high D-waves (worst pull: 2^3D_1 , the $\psi(4160)$).
- Not overfitting — the cross-validation (next) confirms the parameters generalize.

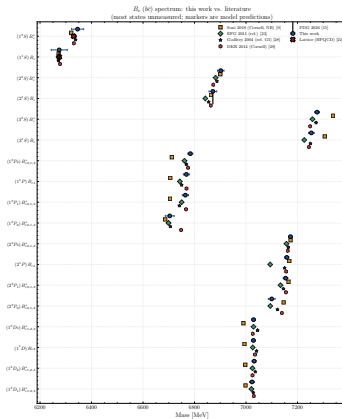
Charmonium vs. the Literature



Per-state deviation $M_{\text{model}} - M_{\text{exp}}$: this work vs. six literature models; green band = ± 10 MeV.

Results: B_c & D — Pure Predictions

- B_c : the two known pseudoscalars are reproduced exactly (the sector is exactly determined); the model then predicts the **entire B_c tower** — S, P and D — dozens of as-yet-unobserved states.
- D : ground state anchored; D^* predicted at 1.985 GeV.
- These mixed-flavour sectors are the real test: *genuine predictions*, consistent with dedicated relativistic models.

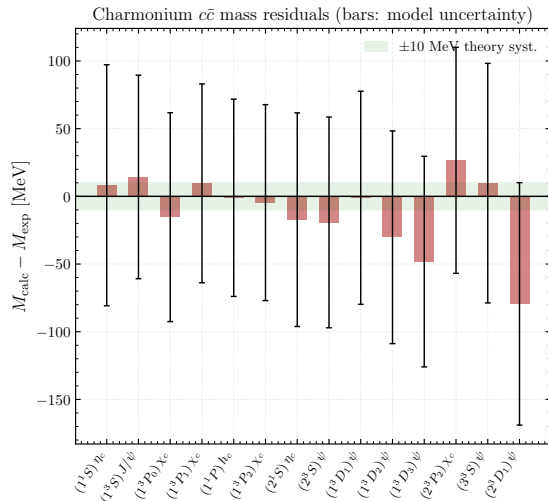
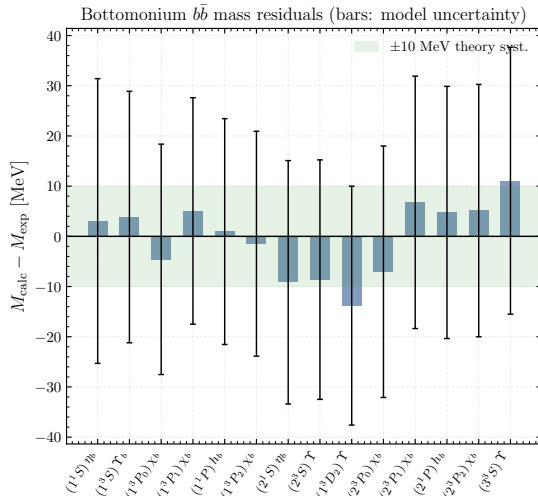


B_c levels: this work vs. literature predictions.

Fit Quality: Residuals & Pulls

- Per-state residuals $M_{\text{calc}} - M_{\text{exp}}$ sit *inside* the ± 10 MeV systematic band for nearly all of bottomonium.
- The handful of charmonium D-wave outliers carry the large pulls — the same relativistic tension seen in the χ^2 .
- Most pulls lie within $\pm 2\sigma$: the error model is honest, not inflated to flatter the fit.

Mass Residuals per State



$M_{\text{calc}} - M_{\text{exp}}$ per state, with model bars and the ± 10 MeV band.

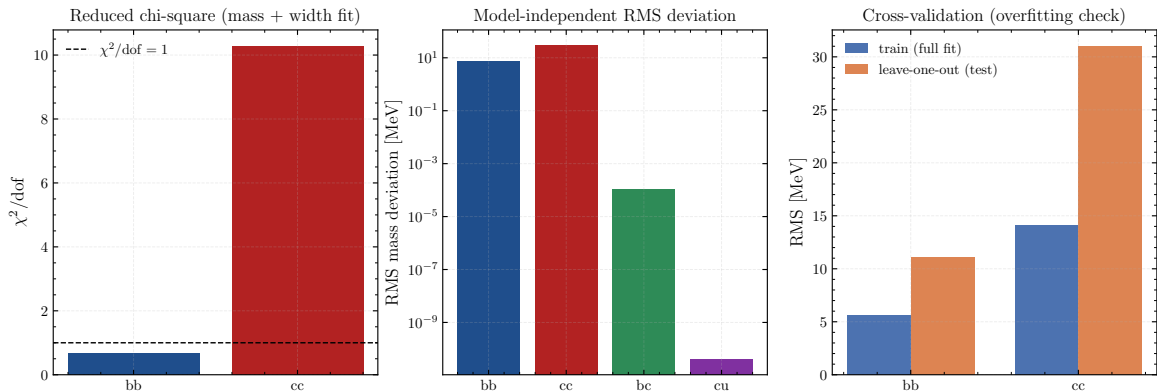
Overfitting Check: Leave-One-Out Cross-Validation

Test: refit the potential on every level *except* one, predict the held-out level, and repeat.

- $b\bar{b}$: LOO 11.1 vs full-fit 5.6 MeV.
- $c\bar{c}$: LOO 31.0 vs full-fit 14.1 MeV.

A ratio ~ 2 with so few levels per free parameter is normal; held-out levels are predicted about as well as fitted ones $\Rightarrow$ the rigid **4-parameter** Cornell form **generalizes** — it is not memorizing the data.

Cross-Validation & Goodness of Fit



χ^2/dof , RMS deviation, and train/test RMS per sector.

6. Results: Decays & Static Properties

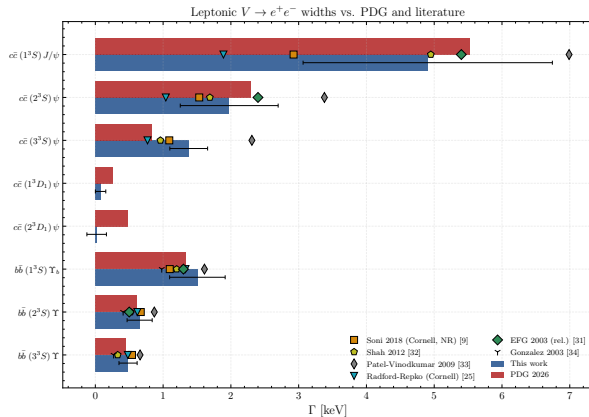
Decay Widths from the Same Wavefunctions

No new parameters. The decay observables are built from the *same* fitted masses and GEM wavefunctions that produced the spectrum — the spectroscopy and the decays are one model, not two.

- **Leptonic** $V \rightarrow e^+ e^-$ — Van Royen–Weisskopf, $\propto |R(0)|^2$.
- **Two-photon** $P \rightarrow \gamma\gamma$ — also $\propto |R(0)|^2$.
- **Radiative** $E1$ and $M1$ transitions between levels.
- **Open-flavour strong decay** via the 3P_0 pair-creation model (e.g. $\psi(3770) \rightarrow D\bar{D}$).

Leptonic Widths: $V \rightarrow e^+e^-$

- Ground and excited Υ and ψ widths agree with PDG at the **10–25%** level.
- $\Upsilon(1S)$: 1.51 vs 1.34 keV; J/ψ : 4.90 vs 5.53 keV.
- The agreement validates $|R(0)|^2$ — the same quantity that feeds every annihilation width.



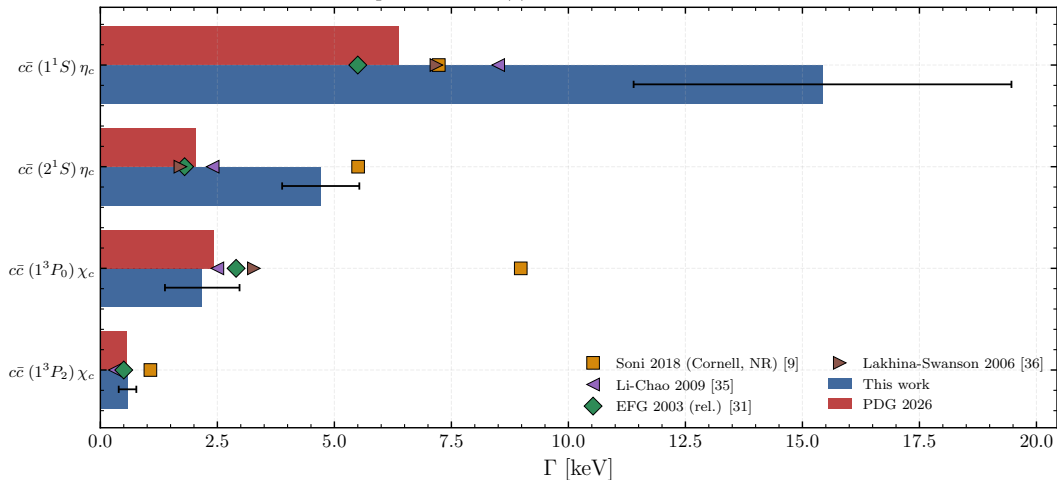
Leptonic $V \rightarrow e^+e^-$ widths, this work vs. PDG.

Two-Photon Widths & an Honest Outlier

- The χ_c and χ_b $\gamma\gamma$ widths come out reasonable; but $\eta_c \rightarrow \gamma\gamma$ **overshoots by $\times 2.4$** (15.4 vs 6.4 keV).
- **Diagnosis:** the non-relativistic *over-concentration* of $|R(0)|^2$ for the light, relativistic charm S-wave — a known NR-limit artefact, *not* a GEM/basis error.
- It is the *same* physics that inflates the charmonium mass χ^2 : the model quantitatively flags its own limit.

Two-Photon Widths

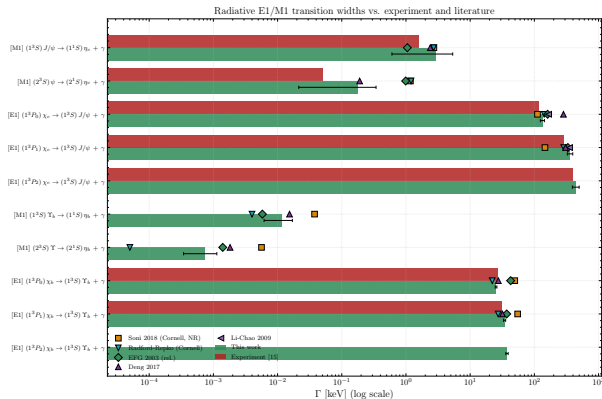
Two-photon $P \rightarrow \gamma\gamma$ widths vs. PDG and literature



Two-photon $P \rightarrow \gamma\gamma$ widths vs. PDG.

Radiative Transitions: $E1$ & $M1$

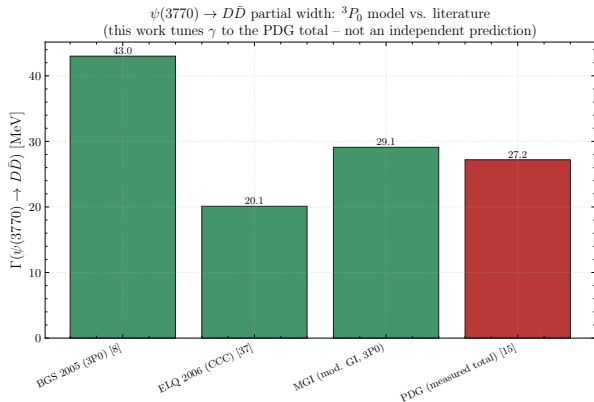
- $E1$ ($\chi_c \rightarrow J/\psi \gamma$, $\chi_b \rightarrow \Upsilon \gamma$) reproduce experiment well — e.g. $\chi_{b2} \rightarrow \Upsilon$ at 37.5 keV.
- $M1$ ($J/\psi \rightarrow \eta_c \gamma$) are tiny and notoriously model-sensitive: order-of-magnitude agreement with a large, honest uncertainty band.
- Together they probe the wavefunction *overlaps* between levels — a stiffer test than the spectrum alone.



$E1/M1$ transition widths vs. experiment (log scale).

Open-Flavour Decay: $\psi(3770) \rightarrow D\bar{D}$

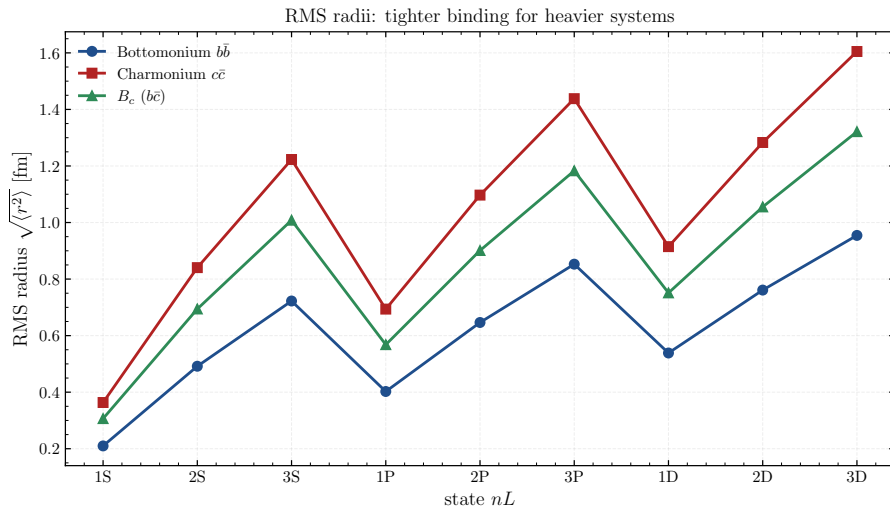
- **3P_0 model:** above the open-charm threshold a light $q\bar{q}$ pair is created from the vacuum with vacuum quantum numbers, fragmenting the quarkonium into a meson pair.
- The $\psi(3770)$ — a $2S$ – $1D$ admixture (hence the S – D mixing earlier) — decays predominantly to $D\bar{D}$.
- The computed strong width is compared directly with the measured value, closing the loop from potential $\rightarrow$ wavefunction $\rightarrow$ hadronic width.



$\psi(3770) \rightarrow D\bar{D}$ via 3P_0 pair creation.

- $\sqrt{\langle r^2 \rangle}$ grows with radial excitation n and is *smaller* for the heavier systems ($b\bar{b} 1S \approx 1.1$ vs $c\bar{c} 1S \approx 2.1 \text{ GeV}^{-1}$).
- Tighter bottomonium binding is exactly what underlies its larger $|R(0)|^2$ and leptonic widths — the picture is internally consistent.
- A free by-product of the same eigenvectors: no extra computation.

RMS Radii per State



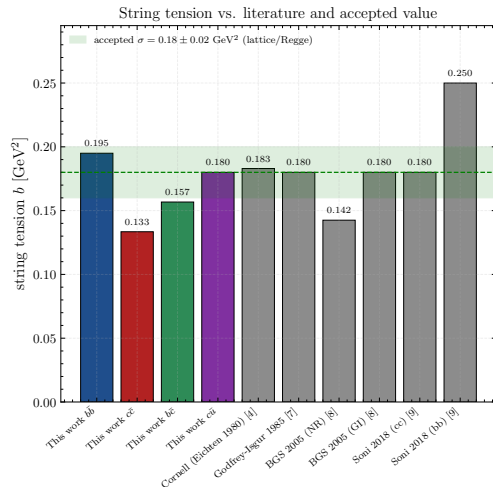
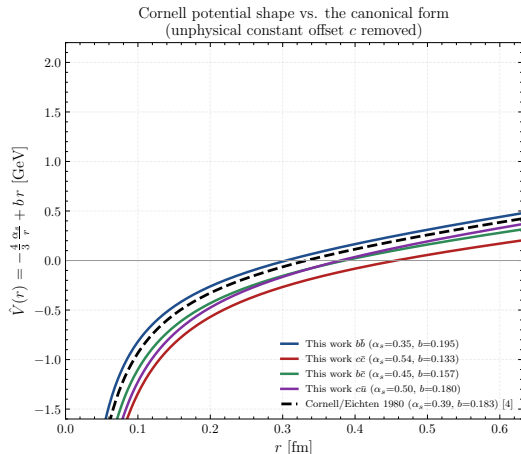
RMS radii $\sqrt{\langle r^2 \rangle}$ per state, per sector.

7. Validation & Outlook

Validation Against the Literature

- **The potential is physical:** the fitted string tension $b \approx 0.13\text{--}0.20 \text{ GeV}^2$ brackets the accepted lattice/Regge value $\sigma \approx 0.18 \text{ GeV}^2$, and the Cornell shape overlays the canonical Eichten 1980 potential.
- **Accuracy among published models:** the aggregate RMS sits with dedicated relativistic models — bearing in mind this work is *fitted* to these masses, so the meaningful comparison is the spread of the genuine-prediction models.

Cornell Shape & String Tension vs. the Literature



Cornell shape vs. Eichten 1980; string tension vs. the accepted $\sigma \approx 0.18 \pm 0.02 \text{ GeV}^2$.

- **One analytic framework:** a closed-form Cornell + GEM solver reproduces bottomonium to **7 MeV** and charmonium to **29 MeV**, cross-validated as *not* overfitting.
- **Spectrum and decays from the same wavefunctions:** leptonic, two-photon, radiative and open-flavour widths — and the lone clear outlier ($\eta_c \gamma\gamma$) *quantitatively* marks the non-relativistic limit rather than a numerical fault.
- **Predictive reach:** the rigid four-parameter form extends to the mixed-flavour B_c and D , predicting **dozens** of yet-unobserved states.

- **Complete the catalogue:** extend the decay-width and RMS-radius tables uniformly across all four sectors and all predicted levels.
- **Refine the charm sector:** add relativistic / coupled-channel corrections to tame the $c\bar{c}$ tension and the $\eta_c \rightarrow \gamma\gamma$ overshoot.
- **Three-body extension:** promote the same closed-form GEM machinery to the **three-body problem** in Jacobi coordinates — opening the door to heavy *baryons*.

One analytic, validated potential model spans $b\bar{b} \rightarrow c\bar{c} \rightarrow B_c \rightarrow D$.